

ARIF SYRAJ

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EDUCATION

University of San Francisco

Master of Science, Computer Science

San Francisco, CA

August 2023 – December 2025

Relevant Coursework: Algorithms, Principles of Software Development, Foundations of A.I., Systems Programming, Software Development Lifecycle, Cloud Computing

University of Illinois at Urbana-Champaign

Bachelor of Science, Mechanical Engineering

Champaign, IL

August 2016 - May 2021

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, JavaScript, HTML5, CSS3, SQL

General: Finetuning, Prompt Engineering, Agile, Data Scraping, Shell Scripting, Containerization, Multithreading, CI/CD, Automated Testing, Cloud Deployment, Network Protocols

Frameworks/Tools: Linux, AWS, GCP, Docker, ROS, pytorch, tensorflow, MongoDB, React, Node.js, Express, JUnit

EXPERIENCE

Software Engineer

EarthSense, Inc.

March 2021 – July 2022

Champaign, IL

- Designed an algorithm to estimate plant height using noisy LiDAR data collected by our phenotyping robot, cutting error by >50% across datasets over the previous algorithm and beating accuracy of manual measurements.
- Revamped robot autonomy by designing a crash detection algorithm fusing LiDAR and odometry data collected from various environments, reducing overall false positives and negatives by 75%.
- Greatly decreased need for manual waypoint recording by implementing automatic waypoint generation on turns, saving customers several minutes per data collection and improving robot autonomy.
- Introduced a culture of documentation in our workflows for libraries and testing methodologies, which I later used to onboard interns for a summer.
- Extensively tested robot system functionality, standardized testing methodologies and created utility shell scripts, helping catch several system bugs before deployment to customers.

Teaching Assistant - AI For Redistricting

University of San Francisco

January 2025 – May 2025

San Francisco, CA

PROJECTS

LLM's To Detect Chart Misinformation

August 2024

- Finetuned a multimodal LLM (LLaVA) to detect visual misinformation in bar charts with LoRA and 2.3k examples.
- Created a custom finetuning script and explored parameters to prevent overfitting and minimize GPU loads.
- Resulted in a model that can identify visually manipulated bar charts 90% of the time.

Highly Available Cloud Application Deployment

October 2024

- Architected and deployed a highly available, secure application on AWS using ECR, EC2, ALB, VPC, and ASGs.
- Configured an ALB to distribute traffic across EC2 instances in private subnets across multiple AZ's with launch templates and auto-scaling groups, enhancing scalability, security and availability.
- Improved operational security by launching bastion hosts across availability zones using ASGs for highly available, secure SSH access to EC2 instances.
- Set up TLS encryption using ACM and Route 53 for domain management and HTTPS-encrypted communication.
- Deployed a containerized application orchestrated with ECS.

Syllabus Generator

May 2024

- Used low-cost LLMs and crewAI to coordinate the automatic generation of a syllabus using basic course info.
- Implemented prompt engineering techniques to optimize syllabus section generation, significantly improving output quality and relevance.
- Developed a microservice using FastAPI that talks to large LLM's like ChatGPT for automated syllabus generation.

AWARDS/PUBLICATIONS

University of San Francisco Computer Science Scholarship - Merit Based Award

May 2023

EnterpriseWorks Top Summer Intern Award

July 2021

- Received a summer intern award in the 'Best Technical Innovation' category by the EnterpriseWorks Incubator.